A. Reconstruct and Challenge Natural-Language Arguments

Core vocabulary and response rule

An **argument** contains **premises** offered in support of a **conclusion**. An **intermediate conclusion** is supported by earlier claims and then supports a later conclusion. A **bridge premise** supplies a missing link. An argument is **valid** exactly when no interpretation makes all premises true and the conclusion false.

On the Core, complete all requested parts in A, but any requested explanation or justification must be only one or two sentences. Each prompt requests at most two compact results; a full argument map is not required.

A.1 Argument structure

Mark indicator words (*because, since, therefore, so*), then ask:

1. What is the author ultimately trying to establish?
2. Which claim is supported and then reused as support?
3. Which claims are evidence, concessions, or background?
4. What bridge would make the last support step reasonable?

An argument map such as $P_1 + P_2 \Rightarrow I$ and $I + P_3 \Rightarrow C$ records convergent or linked support without extra prose.

A.1: similar example

Problem. Read this argument: "Encrypted backups can be restored after device loss. The lab's archives are encrypted backups, so the archives are recoverable. Recoverable records reduce interruption after a failure. Therefore the lab should archive every final data set." State the main conclusion and the intermediate conclusion. No full argument map or extra justification is required.

Worked solution (teaching explanation). The policy recommendation is the endpoint: the other claims explain why the lab should adopt it. The claim that the archives are recoverable is first supported by the first two premises and then reused as support for the recommendation. A fuller teaching map is $P_1 + P_2 \Rightarrow I$, then $I + P_3 +$ bridge $\Rightarrow C$. The policy bridge connects the benefit of recoverability to what the lab should do; identifying that bridge is useful practice, but is not another requested answer.

Answer. Main: archive every final data set. Intermediate: the lab's archives are recoverable.

A.2 Conditionals in English

$$
\begin{aligned}
P \text{ only if } Q &: P \to Q && (Q \text{ necessary for } P),\\
P \text{ if } Q &: Q \to P && (Q \text{ sufficient for } P),\\
P \text{ iff } Q &: (P \to Q) \wedge (Q \to P),\\
P \text{ unless } Q &: \neg Q \to P \quad \text{or} \quad P \vee Q.
\end{aligned}
$$

The **contrapositive** $\neg Q \to \neg P$ is equivalent to $P \to Q$. The converse $Q \to P$ and inverse $\neg P \to \neg Q$ are not.

A.2: similar example

Problem. Let A mean access is granted, T mean training is complete, and S mean a supervisor is present. The rule is: "Access is granted exactly when training is complete and a supervisor is present." Translate the rule and name one condition necessary for A.

Worked solution (teaching explanation). "Exactly when" gives both directions, $A \to (T \wedge S)$ and $(T \wedge S) \to A$. Thus each of T, S is necessary for access, while their conjunction is sufficient. Related translations clarify the distinction: "access only if trained" is $A \to T$; "training and supervision are enough" is $(T \wedge S) \to A$. The contrapositive of $A \to T$ is $\neg T \to \neg A$; the converse $T \to A$ is unsupported because supervision may be absent.

Answer. $A \leftrightarrow (T \wedge S)$; training ($T$) is necessary for A.

A.3 Validity, countermodels, and repair

Valid rules: modus ponens ($P \to Q, P \therefore Q$), modus tollens ($P \to Q, \neg Q \therefore \neg P$), hypothetical syllogism, disjunctive syllogism, conjunction, simplification, and resolution.

For an invalid argument, give a **countermodel**: one assignment making all premises true and the conclusion false. A **minimal repair** adds or changes only what is needed to block that countermodel.

A.3: similar example

Problem. "If the cache is stale, the test fails. The test failed. Therefore the cache is stale." Let S mean the cache is stale and F mean the test fails. State whether the argument is valid; if invalid, give a true-premise/false-conclusion countermodel. One line is enough.

Worked solution (teaching explanation). The form is $S \to F, F \therefore S$, which affirms the consequent. Modus ponens would require S as a premise. When $S = \text{F}$ and $F = \text{T}$, the conditional and the second premise are true, but the conclusion is false. For additional practice, adding $F \to S$ (or strengthening the first premise to $S \leftrightarrow F$) repairs the inference. Repair is not requested here.

Answer. Invalid; $S = \text{F}$, $F = \text{T}$.

Frequent A errors

Do not reverse "only if." Do not call evidence the main conclusion. A countermodel must make *every* premise true and the conclusion false. A repair should close the identified gap, not merely restate the conclusion.

A readiness

I can map premises, intermediate conclusions, and the main conclusion; translate necessary and sufficient conditions; use contraposition without using a converse; and test validity with a rule or a true-premise/false-conclusion countermodel.

B. Logical Reasoning I: Anatomy and Inference

How B is graded

For every B problem, choose one answer only. **No written justification is required.** The worked explanations here teach a repeatable selection method.

B.1 Main conclusion

The **main conclusion** is the claim the author ultimately wants accepted. Ask "Why believe this?" Evidence answers that question; an intermediate conclusion is both supported and reused. A concession often begins with *although* or *admittedly*.

B.1: similar example

Problem. "Weekday appointments occur when many hourly workers cannot attend. Recent evening clinics filled immediately. A nurse and funding are now available. Therefore the clinic should offer Saturday appointments." Which statement is the main conclusion? Choose one answer; no written justification is required.

(a) Weekday appointments are inconvenient for many hourly workers.
(b) Recent evening clinics filled immediately.
(c) A nurse is now available.
(d) The clinic should offer Saturday appointments.
(e) Funding for the clinic is now available.

Worked solution (teaching explanation). Ask which claim the other sentences answer "Why believe that?" Inconvenient weekday timing and full evening clinics support demand; the nurse and funding make the proposal feasible. All three support the Saturday recommendation.
Answer. (d).

B.2 Role of a statement

Describe what the statement *does*: premise, intermediate conclusion, main conclusion, concession, objection, reply, example, or background. Read one sentence before and after it and identify the claim it supports or qualifies.

B.2: similar example

Problem. "Digitizing the archive will interrupt access for two weeks. However, it prevents irreversible loss, and the interruption can be reduced by scanning in batches. The archive should be digitized." What role does the first sentence play? Choose one answer; no written justification is required.

(a) It states the main conclusion.
(b) It concedes a cost that the author answers while defending the recommendation.
(c) It supplies an example proving that digitization prevents loss.
(d) It is an intermediate conclusion supporting the final recommendation.
(e) It states a principle that rules out digitization.

Worked solution (teaching explanation). The last sentence is the recommendation. The first counts against it; "however" introduces reasons the author thinks outweigh that cost. Thus the author acknowledges an objection and responds to it.
Answer. (b).

B.3 Point at issue

Find one exact proposition P such that one speaker accepts P and the other accepts $\neg P$. Agreement about background facts is not disagreement about the inference drawn from them.

B.3: similar example

Problem. Mina: "A random sample of twelve is enough to estimate majority support." Omar: "Random selection helps, but twelve is too uncertain to determine a majority." Which proposition is a point at issue between them? Choose one answer; no written justification is required.

(a) Random selection can improve a sample.
(b) The sample contains twelve people.
(c) A random sample of twelve is sufficient to determine majority support.
(d) Every larger sample is perfectly accurate.
(e) Majority support can never be estimated.

Worked solution (teaching explanation). Both can accept that random selection helps. Mina accepts the sample-size sufficiency claim and Omar rejects it. A disagreement must concern the exact same proposition, not an unstated belief about polls in general.
Answer. (c).

B.4 Must follow

A correct answer is true in *every* situation satisfying the facts. Chain conditionals, apply contrapositives, and respect quantifiers. Reject choices that are merely possible, likely, or stronger than the premises.

B.4: similar example

Problem. Every certified device was inspected. No inspected device lacks a seal. Unit K is certified. Which statement must follow? Choose one answer; no written justification is required.

(a) Unit K has a seal.
(b) Every sealed device is certified.
(c) Every inspected device is certified.
(d) Every device is inspected.
(e) Unit K is the only certified device.

Worked solution (teaching explanation). Certification implies inspection, and inspection implies having a seal. Apply each implication to K to conclude that K has a seal. The reverse implications are not supplied; the facts also say nothing about every other device.

Answer. (a).

B.5 Parallel reasoning

Abstract the structure before comparing topics. Preserve arrow direction, negations, conjunction/disjunction, and quantifier strength.

B.5: similar example

Problem. The argument has form $P \to Q$, $P \vee R$, $\neg Q$, therefore R. Which argument has the same reasoning pattern? Choose one answer; no written justification is required.

(a) If the pump runs, water flows. Water flows. Therefore the pump runs.
(b) If water flows, the pump runs. The pump runs or the valve is closed. No water flows. Therefore the valve is closed.
(c) If the pump runs, water flows. The pump does not run or the valve is closed. No water flows. Therefore the valve is closed.
(d) If the pump runs, water flows. The pump runs or the valve is closed. Water flows. Therefore the valve is closed.
(e) If the pump runs, water flows. The pump runs or the valve is closed. No water flows. Therefore the valve is closed.

Worked solution (teaching explanation). First use modus tollens on $P \to Q$ and $\neg Q$ to obtain $\neg P$. Then disjunctive syllogism on $P \vee R$ and $\neg P$ yields R. In (e), pump running is P, water flowing is Q, and valve closed is R. The other choices change a premise or arrow direction.

Answer. (e).

Useful distinctions

- **Entailed**: cannot be false when the premises are true.
- **Supported**: made more credible, but not guaranteed.
- **Consistent**: can be true together; neither must imply the other.
- **Scope**: words such as *all, some, most, only, unless* control how strong a claim is.

Frequent B errors

Do not choose a vivid detail instead of the conclusion. For point-at-issue questions, both speakers must take opposite positions on the exact claim. For must-follow questions, do not add ordinary-world assumptions. For parallel reasoning, match form rather than subject matter.

B readiness

I can distinguish conclusion, evidence, concession, and role; isolate a precise disagreement; identify what must follow; and match an argument's logical form. On the Core I record only the selected answer.

C. Logical Reasoning II: Assumptions and Evidence

How C is graded

For every C problem, choose one answer only. **No written justification is required.** Use the tests below privately to select the answer.

C.1 Necessary assumption

A necessary assumption is a condition the argument *requires*. Use the **negation test**: negate a candidate precisely. If the reasoning collapses, the candidate was necessary. The assumption need not prove the conclusion.

C.1: similar example

Problem. "Encrypted backups can restore any file included in the latest completed backup. Therefore every current project file could be restored tonight using that backup." Which assumption does this reasoning require? Choose one answer; no written justification is required.

(a) Every restored file was created today.
(b) Every current project file is included in the latest completed backup.
(c) Encryption makes all backups smaller.
(d) No project file has ever been deleted.
(e) The backup is the only possible way to recover a file.

Worked solution (teaching explanation). The premise covers files in the backup; the conclusion covers every current file. Negate (b): at least one current file is absent. Then the stated backup capability does not cover that file, so this reasoning fails. The conclusion could still hold through another recovery method, but that would be a different argument.
Answer. (b).

C.2 Sufficient assumption

A sufficient assumption, when added, makes the conclusion follow. Test by inserting the choice into the premises and completing a valid chain. It may be stronger than necessary, but it must not leave a gap.

C.2: similar example

Problem. Plan K is the only plan meeting both the budget and the deadline. The committee concludes that it should choose K. Which added principle makes that conclusion follow? Choose one answer; no written justification is required.

(a) Any chosen plan must meet both constraints.
(b) The committee should prefer a low cost even when the deadline is missed.
(c) Every plan meets at least one constraint.
(d) Whenever at least one plan meets both constraints, the committee should choose a plan that meets both.
(e) The committee may postpone making any choice indefinitely.

Worked solution (teaching explanation). The facts identify the unique feasible plan, but do not yet require choosing a feasible plan. Add (d): some plan meets both constraints, so the committee should choose one that does. K is the only such plan, forcing the choice of K. A rule merely forbidding infeasible choices would not require making any choice.
Answer. (d).

C.3 Diagnose the flaw

State the gap specifically. Common flaws include converse error; confusing correlation with causation; unrepresentative sample; circular reasoning; part-to- whole or whole-to-part error; false dilemma; equivocation; and treating lack of evidence as evidence of absence.

C.3: similar example

Problem. "The park installed benches in May. Attendance rose afterward. Therefore the benches caused the increase in attendance." Which flaw best describes this argument? Choose one answer; no written justification is required.

(a) It infers causation from an uncontrolled before–after association.
(b) It proves the conclusion and then denies it.
(c) It assumes that no one ever uses benches.
(d) It rejects a claim solely because of its source.
(e) It mistakes a statement about every park for one about a single park.

Worked solution (teaching explanation). The evidence gives a before–after association. Weather, programming, or another contemporaneous change could explain the attendance rise. Without an appropriate comparison or control, temporal sequence alone does not establish the proposed cause.
Answer. (a).

C.4 Strengthen

Find the conclusion and its weakest support link. A strengthener may rule out an alternative explanation, validate a measurement, make a sample representative, establish a missing connection, or show the proposed mechanism operates. It need not prove the conclusion.

C.4: similar example

Problem. One office made fewer sorting errors after adopting high-contrast labels. Its manager concludes that the new labels reduce sorting errors. Which finding most strengthens the reasoning? Choose one answer; no written justification is required.

(a) Employees say the new labels look more attractive.
(b) The new labels cost less than the old ones.
(c) Comparable sorting lines were randomly assigned old or high-contrast labels, and only the latter showed a sustained error reduction.
(d) The office changed its labels at the beginning of a month.
(e) Some errors occurred before either label design was introduced.

Worked solution (teaching explanation). The weak link is whether a different change explains the drop. Random assignment and a comparison using old labels help hold other factors comparable. A sustained reduction confined to the high-contrast condition directly supports the proposed mechanism; popularity or lower cost does not.
Answer. (c).

C.5 Weaken

A weakener damages the inference, not necessarily the conclusion. Look for a confounder, selection effect, bad proxy, reversed causation, missing necessary step, or evidence that the mechanism will not operate.

C.5: similar example

Problem. "The application tutorial is accurate. If applicants are required to open it before submitting a form, they will complete their forms correctly." Which fact most weakens this reasoning? Choose one answer; no written justification is required.

(a) The tutorial contains screenshots of the form.
(b) Applicants can reread the tutorial if they wish.
(c) The tutorial is written in the same language as the form.
(d) The tutorial explains all required fields accurately.
(e) Applicants can close the tutorial immediately without reading it.

Worked solution (teaching explanation). The mechanism needs opening the tutorial to lead to reading, understanding, and correct use. Choice (e) breaks the first link: merely opening an accurate resource does not guarantee exposure to its content.
Answer. (e).

Necessary versus sufficient

A necessary assumption N is required by the *argument's support link*: negating N undermines that reasoning. This need not make the conclusion false; it might be true for a different reason. A sufficient assumption S supplies enough for C: premises plus S imply C. Do not confuse a condition needed by this argument with a condition needed for the conclusion to be true at all.

Frequent C errors

The negation of "all" is "at least one not," not "none." A necessary assumption can be modest; a sufficient assumption can be strong. Strengtheners and weakeners target the support link, not whether you personally agree with the conclusion. Name the passage-specific flaw, not only a memorized label.

C readiness

I can apply the negation test; distinguish necessary from sufficient assumptions; name a precise inferential gap; and choose evidence that changes the actual support link. On the Core I record only the selected answer.

D. Principles and Explanations; E. Evaluation and Revision

How D and E are graded

In D, choose one answer only; **no written justification is required**. In E, complete every requested part, but keep any requested explanation or justification to **one or two sentences**. Concise labels and bullets count.

D.1 Principle support

Translate each principle as conditions → judgment. Match *every* operative condition to a stated fact; reject rules that are too broad, too narrow, or point toward the opposite judgment.

D.1: similar example

Problem. Mira took every reasonable step to file on time, but an agency outage prevented timely filing; logs establish both facts. A reviewer concludes that her late fee may be waived. Which principle most directly supports that conclusion? Choose one answer; no written justification is required.

(a) A fee may be waived only when no filing attempt was made.
(b) A fee may be waived when reasonable compliance was prevented by circumstances outside the applicant's control.
(c) Every applicant who misses a deadline must receive a waiver.
(d) An agency outage always makes an applicant responsible for the delay.
(e) A fee may never be waived after a deadline.

Worked solution (teaching explanation). The matching principle must connect both reasonable effort and the external outage to permission to waive the fee. Choice (b) includes both conditions and reaches exactly that judgment. The other choices omit, reverse, or deny the relevant connection. **Answer.** (b).

D.2 Resolve the discrepancy

Treat both observations as true. Find a third fact that separates the measures, groups, or time periods so the apparent conflict disappears.

D.2: similar example

Problem. A workshop had fewer registered participants this year than last year, yet more seats were occupied at the meeting. Each attendee occupied one seat, and no unregistered guests attended. Which fact best resolves the apparent discrepancy? Choose one answer; no written justification is required.

(a) The workshop used the same registration system.
(b) Every registered participant attended in both years.
(c) This year's no-show rate was much higher.
(d) This year's registered group had a much lower no-show rate.
(e) The meeting had fewer occupied seats this year.

Worked solution (teaching explanation). Registration and actual attendance are different measures. Suppose last year 100 registered and 50 attended, while this year 80 registered and 72 attended. The lower no-show rate produces fewer registrations but more occupied seats. Choice (d) permits exactly this pattern without denying either observation. **Answer.** (d).

D.3 Parallel flaw: similar example

Problem. "Every certified device has been inspected. This device has been inspected. Therefore it is certified." Which argument has the same flawed form? Choose one answer; no written justification is required.

(a) Every oak is a tree. This plant is an oak. Therefore it is a tree.
(b) Every oak is a tree. This plant is a tree. Therefore it is an oak.
(c) Every oak is a tree. This plant is not a tree. Therefore it is not an oak.
(d) Every oak is a tree. This plant is not an oak. Therefore it is not a tree.
(e) Some trees are oaks. This plant is an oak. Therefore some plants are trees.

Worked solution (teaching explanation). The original has form $P \to Q, Q \therefore P$: it affirms the consequent. Choice (b) repeats that exact error. Choice (d) is also invalid but denies the antecedent instead. Match the logical structure rather than merely choosing another dubious conclusion. **Answer.** (b).

D.4 Method of reasoning

Describe the argument's actual operation: applies a rule, eliminates alternatives, uses an analogy, gives a counterexample, distinguishes cases, concedes then responds, or shows that a proposal defeats its own goal.

D.4: similar example

Problem. One speaker claims that deadlines are never fair. An author replies: "The deadline for emergency staffing is fair: every qualified volunteer has a reasonable chance to respond, and a firm cutoff is needed to coordinate the rescue teams. So the claim that deadlines are never fair is false." Which method does the author use? Choose one answer; no written justification is required.

(a) Giving a counterexample to a universal claim.
(b) Showing that every deadline is fair.
(c) Rejecting the speaker because of a personal defect.
(d) Inferring fairness from popularity alone.
(e) Assuming that no deadline can coordinate staffing.

Worked solution (teaching explanation). "Never fair" is a universal claim. A single fair deadline is enough to refute it. The author supplies such a case; the response does not establish that all deadlines are fair.

Answer. (a).

D.5 Complete the argument

Insert a claim that connects established facts to the stated conclusion. Prefer the weakest answer that closes the exact gap; avoid a new topic or an unnecessary universal rule.

D.5: similar example

Problem. A transit survey samples only early riders. Later riders face systematically different service conditions. The author argues: "Because _____, this survey cannot justify system-wide service cuts." Which completion best connects the facts to the conclusion? Choose one answer; no written justification is required.

(a) every early rider favors service cuts
(b) later riders always travel without paying
(c) a sample restricted to a systematically different subgroup cannot support a system-wide estimate
(d) any transit survey proves that all routes are identical
(e) cutting service is always wrong regardless of evidence

Worked solution (teaching explanation). The conclusion concerns the whole system, but the evidence represents one systematically different subgroup. Choice (c) states the relevant limitation of that sampling method. It targets scope without adding an unrelated policy claim.

Answer. (c).

E.1 Causal evaluation

Separate **association** from **causation**. Name the conclusion, evidence, and a plausible confounder, selection effect, reverse cause, or measurement problem. Strong evidence uses random assignment, a valid comparison group, baseline measures, or converging mechanisms.

E.1: similar example

Problem. Students who enabled submission reminders submitted assignments on time more often than students who did not. An administrator concludes that reminders cause timely submission. Give one plausible alternative explanation and one study design that would help test the causal claim. Keep the response to two brief sentences.

Worked solution (teaching explanation). This is observational: students selected whether to enable reminders. Already-organized students may both turn on reminders and submit on time. Random assignment of comparable students helps separate that pre-existing difference from the effect of reminders.

Answer. Already-organized students may be more likely both to enable reminders and to submit on time. Randomly assign comparable students to reminders or no reminders and compare timely-submission rates.

E.2 Apply a principle

List the rule's conditions, compare each with the case, then state the result. Do not silently add an exception. If a term is ambiguous, state one reasonable interpretation in a short sentence.

E.2: similar example

Problem. Principle: "A penalty based on a screening method with a high false-positive rate is justified only if reliable independent evidence corroborates the result." A student's screen is positive, but no independent corroborating evidence is available. Does the principle justify imposing the penalty now? Give the result and one short sentence identifying the decisive condition.

Worked solution (teaching explanation). The rule makes corroboration necessary for a justified penalty. Here that condition is absent, so the rule does not justify the penalty. The screen may justify a proportionate investigation, but that is a different action from imposing a penalty.

Answer. No. Reliable independent corroboration, a required condition for the penalty, is absent.

E.3 Integrated Op-Ed analysis and revision

The requested response has two parts: state the main conclusion, then revise it to the strength the evidence warrants. To prepare that short answer, identify the strongest support and its main gap. A good revision preserves useful evidence while adding scope, comparison, or uncertainty.

E.3: similar example

Problem. "One bus-lane pilot reduced travel time on its corridor and coincided with higher ridership. Therefore the city should convert every city road to a bus lane." State the main conclusion, then give one evidence-supported revision. Two compact sentences are enough.

Worked solution (teaching explanation). The main conclusion recommends converting every road. Evidence from one corridor does not establish effects on every road, and the observed ridership change may have another cause. A cautious revision preserves the pilot's encouraging evidence while limiting the recommendation to further testing on comparable corridors.

Answer. Main conclusion: convert every city road to a bus lane. Revision: test bus lanes on comparable corridors and measure travel time, safety, and spillovers before deciding whether broader conversion is warranted.

Frequent D–E errors

Do not choose a principle missing one stated condition. Resolve a discrepancy without denying either observation. Match a flaw's form, not its topic. In E, do not write an essay: identify the decisive gap and the evidence or revision that addresses it in one or two sentences.

Final readiness check

- A: I can reconstruct, translate, test, countermodel, and repair.
- B–D: I can select the answer using the named test; I know no written justification is required.
- E: I can evaluate causation, apply a principle, or revise an argument in a complete but compact one- or two-sentence response.
- I can distinguish what is entailed, merely supported, and merely possible.